

# Introduction to Artificial Intelligence

Basics, fundamentals, history and context

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INFO-02020

September 2026

# What is (Artificial) Intelligence?

Disclaimer: The concept of intelligence is not easily defined and it is still discussed till this day.

**Intelligence** is the capacity of a system to use information currently held in it to adapt to the requirements of the context and behave in such a way so that the system achieves a given goal.

Throughout history, many aspects have been theorized around it:

- Psychometric: General factor (Spearman) vs Composed (Thurstone), Hierarchical models (Ej. Carroll) and Non-hierarchical models (Guilford)
- Cognitive-Components (Sternberg): information-processing components (Triarchic theory).
- Multiple units of intelligence (Gardner) -Although he has been hardly criticized by his (lack of) methods.
- Biologically and Evolutionary-based approaches: origins and adaptation of organs (brain regions) and information processing capacities.

**Artificial Intelligence** (AI) refers to a field which studies and develops models implementables by machines that provides "human-like" capacities to behave and process rationally in order to solve a task

# Human-like processing?

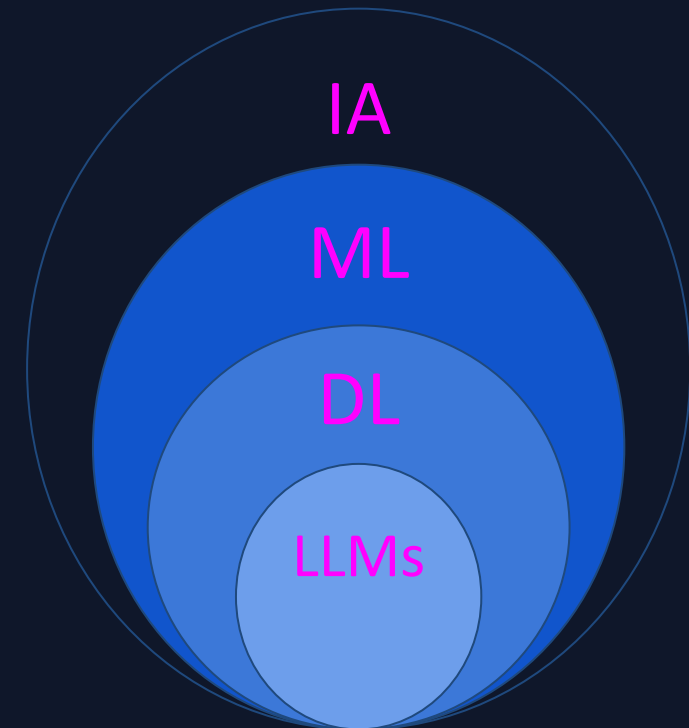
|          | Humanly                                                                                                                                                                                                                                                                                                                 | Rationally                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acting   | <p>Turing Test (TT)</p> <ul style="list-style-type: none"><li>• Nat Lang Processing</li><li>• Knowledge representation</li><li>• Automated reasoning</li><li>• Machine Learning (learn -&gt; generalize)</li></ul> <p>Total TT</p> <ul style="list-style-type: none"><li>• + Comp Vision</li><li>• + Robotics</li></ul> | <p>Rationally behaving agent &gt;&gt; get the best outcome out of a behaviour.</p> <p> This is the most successful model &gt;&gt; called the <i>standard model</i></p> |
| Thinking | <p>Process information in a way that resembles how the human mind does it &gt;&gt; better pictured, resembling our models of how the human mind processes information.</p>                                                                                                                                              | <p>Logic and probabilities.</p> <p>Reasoning adequately when facing certainties and uncertainties.</p>                                                                                                                                                    |

Unlike traditional programming where rules and data are fed to get answers, AI (specifically, ML) involves feeding data and answers to discover the underlying rules >> Generalization (a key characteristic of Intelligence)

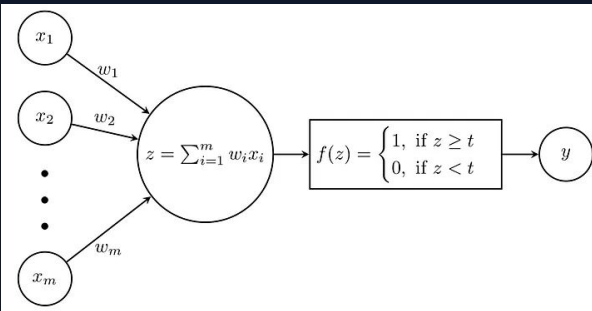
# What is Machine Learning?

**Machine Learning** (ML) is a subfield of AI that focuses on developing models that can learn patterns from data (aka training) to then generalize what has previously learned to face novel, unseen tasks.

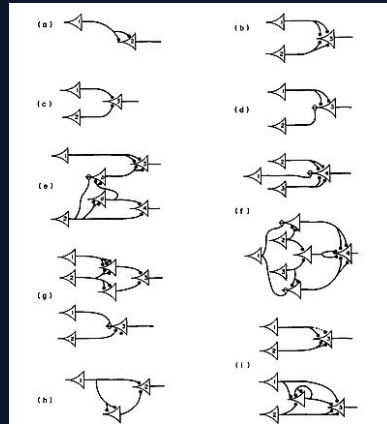
Other AI models may not rely on learning to solve tasks (e.g. logic based models, such as those based on pre-programmed decision trees).



# The Evolution of AI



[Source](#)



## 1939–1952 - the squid

Hodgkin and Huxley combined biology and physics to create a circuit-like model (and its equations) of action potential

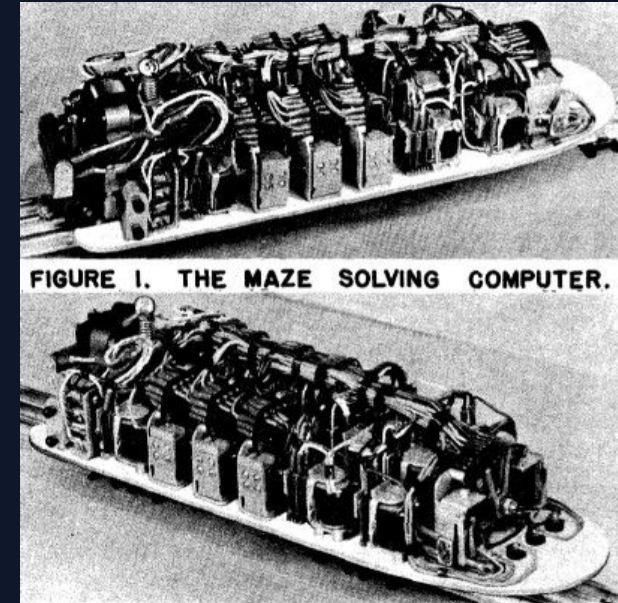
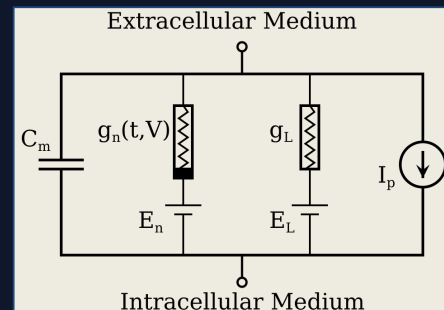


FIGURE 1. THE MAZE SOLVING COMPUTER.

[Source](#)

## 1943 - Cambridge

McCulloch and Pitts established the relation between neurons and boolean logic.



**A QUANTITATIVE DESCRIPTION OF MEMBRANE CURRENT AND ITS APPLICATION TO CONDUCTION AND EXCITATION IN NERVE**

BY A. L. HODGKIN AND A. F. HUXLEY

*From the Physiological Laboratory, University of Cambridge*

*(Received 10 March 1952)*

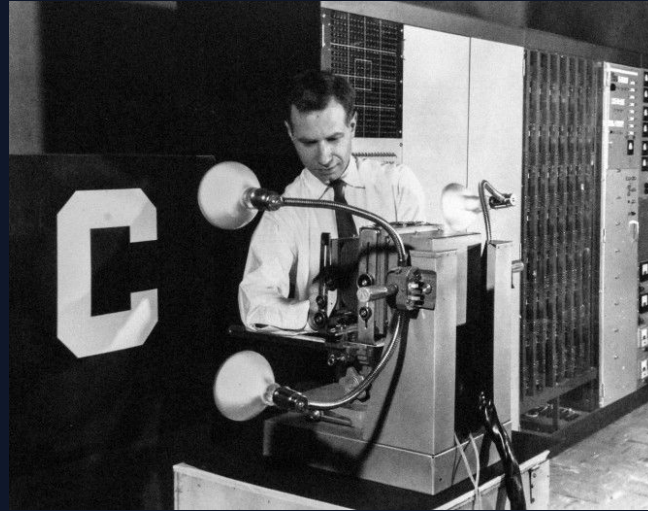
## 1950 - First Neural Net Computer

Minsky and Edmonds (1950) developed the first computer with neuromorphic features, inspired by McCulloch and Pitts

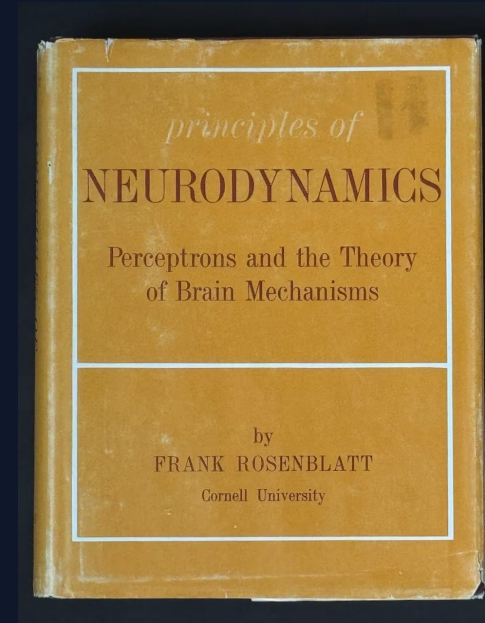
# The Evolution of AI

## The birth of AI as a discipline - 1956

A summer camp that is  
commonly considered the  
formal origin of AI as a  
discipline



Source: Wikimedia Commons



## Rosenblatt's improvements to the Perceptron

He added the model's bias,  
improved weights and  
developed a learning  
algorithm (1958-1962).

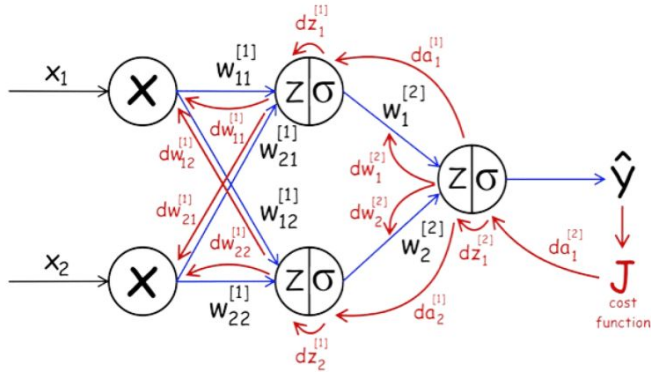
AI Magazine Volume 27 Number 4 (2006) (© AAAI)

## A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence

August 31, 1955

John McCarthy, Marvin L. Minsky,  
Nathaniel Rochester,  
and Claude E. Shannon

# The Evolution of AI

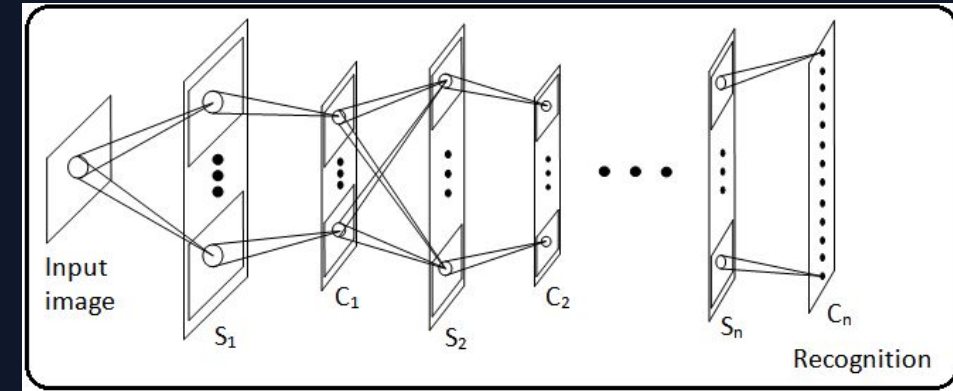


**Backpropagation**

**1982-1984 -**

## **Associative Memories**

Applied "strong" physics (dynamics, attractors) to develop recurrent memories capable of content-addressable memories



## **1986 - Learning and backprop**

Rumelhart, Williams, Hinton popularized backprop as a training algorithm MLPs

*Proc. Natl. Acad. Sci. USA*  
Vol. 79, pp. 2554-2558, April 1982  
Biophysics

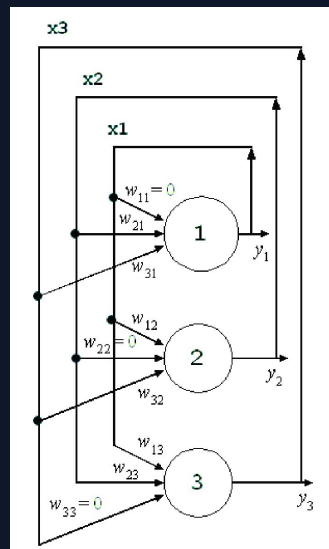
### **Neural networks and physical systems with emergent collective computational abilities**

(associative memory/parallel processing/categorization/content-addressable memory/fail-safe devices)

J. J. HOPFIELD

Division of Chemistry and Biology, California Institute of Technology, Pasadena, California 91125; and Bell Laboratories, Murray Hill, New Jersey 07974

Contributed by John J. Hopfield, January 15, 1982



## **1980- The neocognitron**

Frist CNN for machine vision

# The Evolution of AI

## A fast learning algorithm for deep belief nets \*

**Geoffrey E. Hinton and Simon Osindero**  
Department of Computer Science University of Toronto  
10 Kings College Road  
Toronto, Canada M5S 3G4  
{hinton, osindero}@cs.toronto.edu

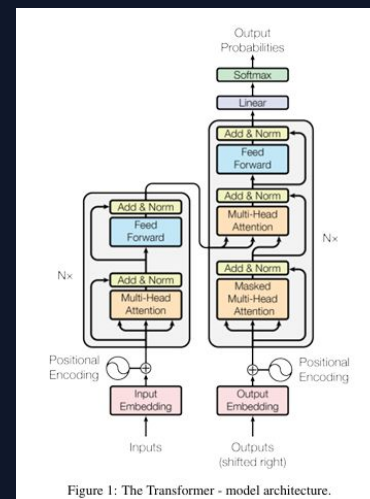
**Yee-Whye Teh**  
Department of Computer Science  
National University of Singapore  
3 Science Drive 3, Singapore, 117543  
tehyw@comp.nus.edu.sg

## 2017 - Attention Is All You Need

Vaswani et al. proposed the concept of transformers, which revolutionized the architectures of LLMs, increasing efficiency and applications

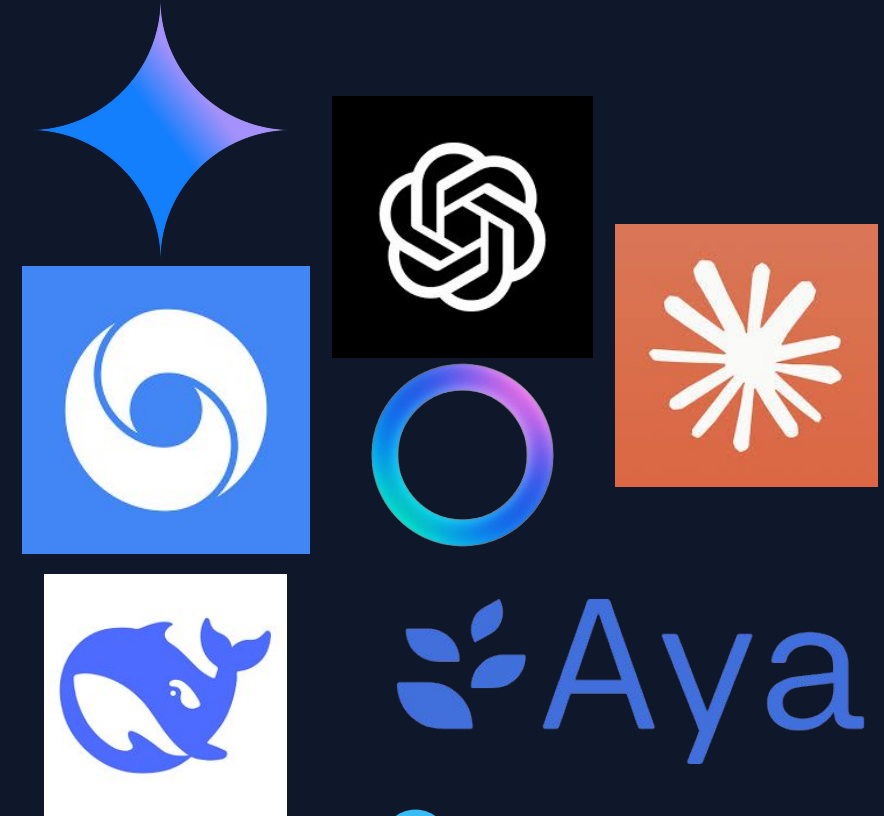
## 2006 - Deep Beliefs Nets

Hinton and his team developed the concept of Deep (Big) NNs



## 2010 to present

Increasing number of companies developing implementations of AI algorithms in different fields



# Applications

## Life Sciences

Alpha Fold: Analysis of protein molecules.

Comp Neuro & NeuroAI: study of brain and mind, as well as inspiration of improvements from them to AI

## Generative AI

Evolution of DNN that can now generate more than text: Images, videos, music and complex combinations between them (Ex. marketing media with Pomelli)

## Transportation

New implementations in vehicles and transportation that provide safety measures, easy travelling and may even automate driving.

# Applications

## Automatization

They can be used from learning tools useful to interact with the environments to learn complex sequences and patterns.

## Control systems

For systems where control is useful, AI can be used to develop measures to provide control over complex systems and achieve a desired goal state.

## Cybersecurity

From analysing vast amounts of data in real time, to developing agents that can hack or confronts vulnerabilities

# Applications

## **Social media**

Recommending content and suggesting features through analysis of general and individual patterns within and across platforms

## **Retail**

Storing and analysing customer and consumer data in order to recommend products and services at the best time.

## **Agentic services and integration**

Facilitate integration of agents in daily routines of people: from scheduling meetings, to arranging info for daily digests, execute home chores or purchases.

# Some important (work) considerations

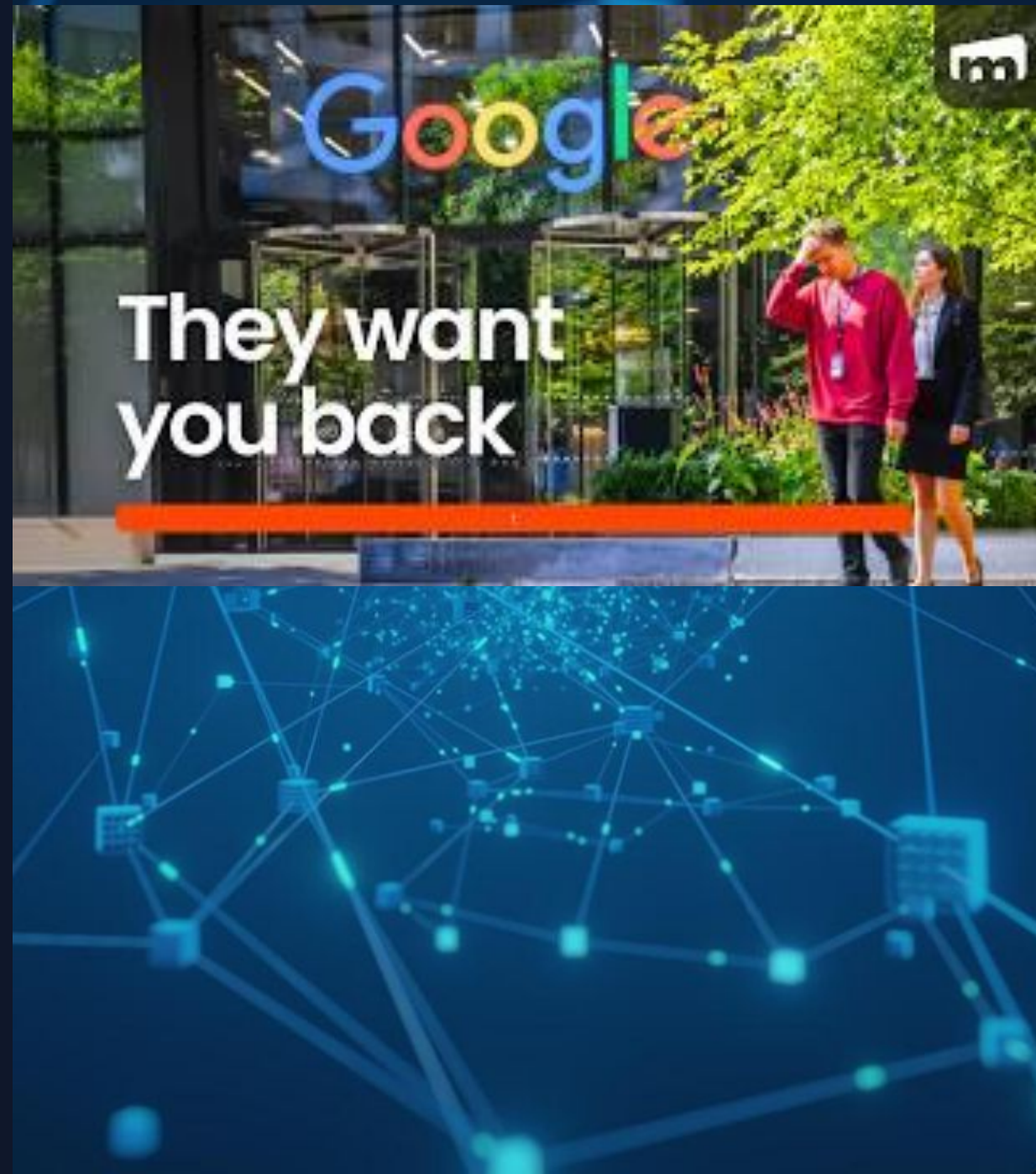
AI remains a good tool, but just a tool and not a complete replacement of human labour.

As of today, we do require humans to complete tasks all over society.

Main Adversely affected:

1. Recent graduates, Jr Positions, Entry-level (Mahoney, McEntarfer, Wahal, 2026).
2. Execution-based positions vs strategic ones.

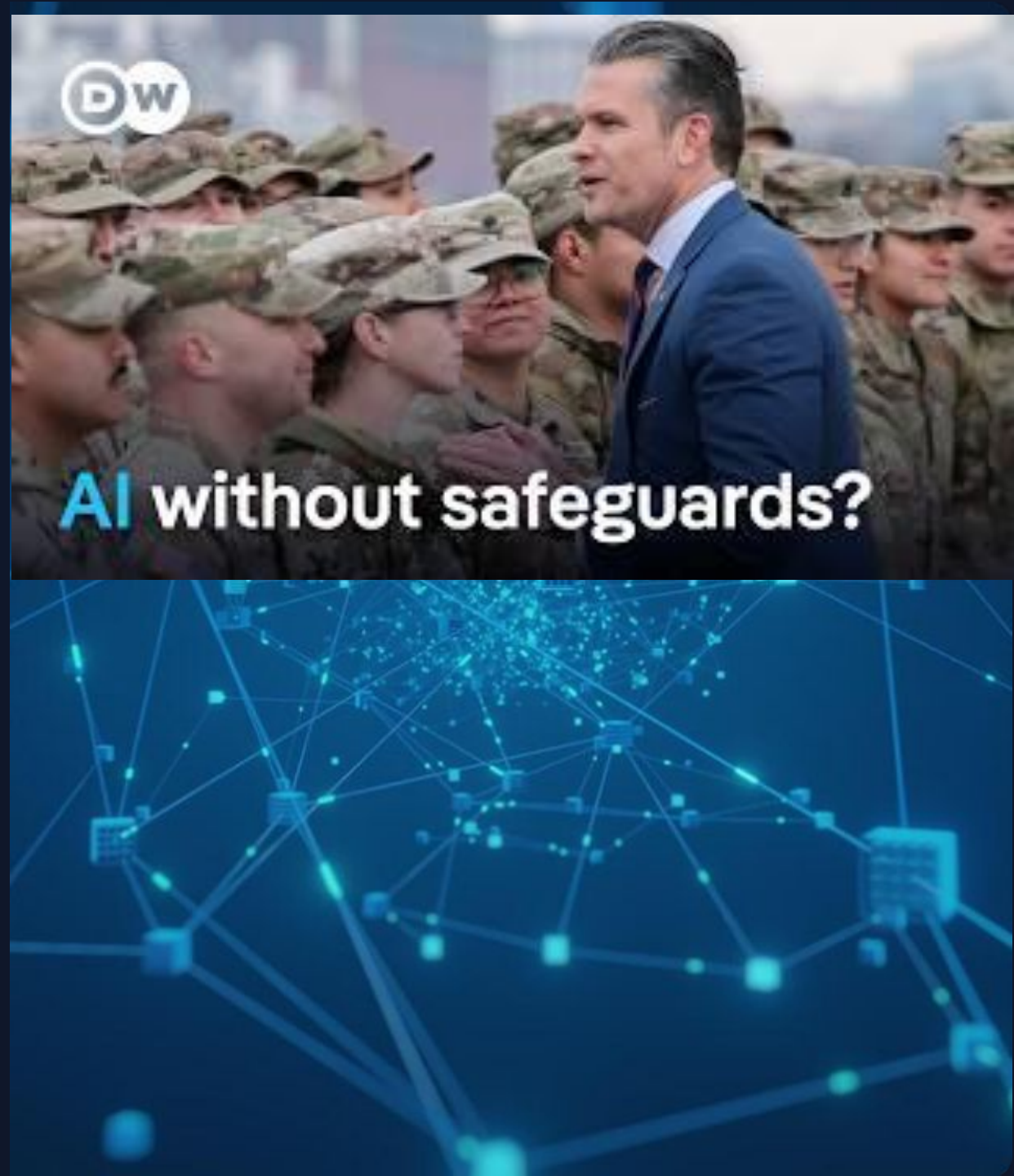
Initial AI Hype-based laying off seems to have been "too deep", thus rehiring projects itself to be high (and costly)



# Some important (defense) considerations

AI is a tool, a popular one. There are high hopes for what it can do and how much money there is to gain from it.

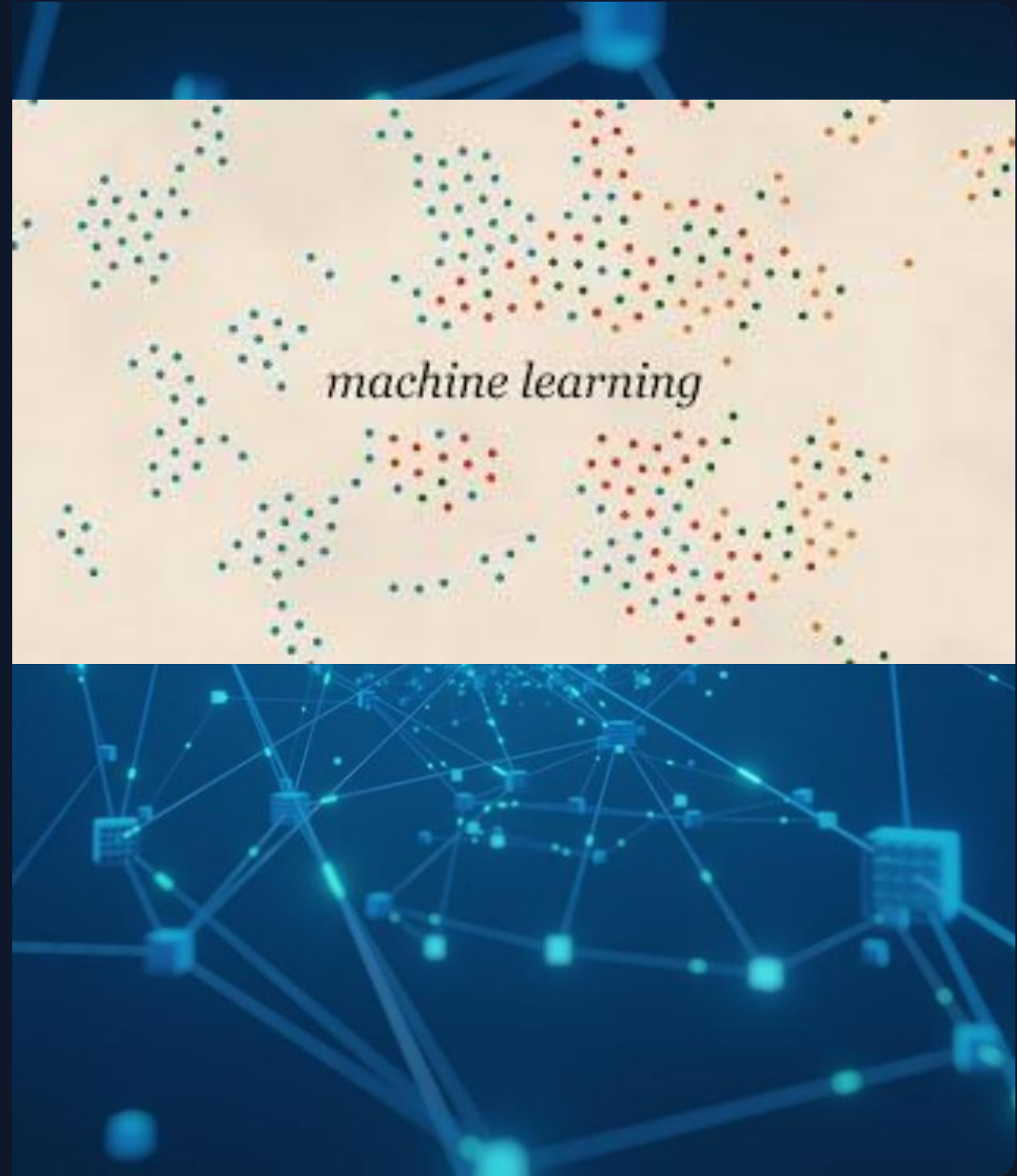
As of today, AI seems to advance at a rhythm that outpaces current legislation. So one must remain ethically aware of these (and more) aspects when training, studying or deploying AI based products and services to the real world.



# Some important (bias) considerations

AI is a tool, a popular one. There are high hopes for what it can do and how much money there is to gain from it.

As of today, AI seems to advance at a rhythm that outpaces current legislation. So one must remain ethically aware of these (and more) aspects when training, studying or deploying AI based products and services to the real world.



# Some important (surveillance) considerations

AI is a tool, a popular one. There are high hopes for what it can do and how much money there is to gain from it.

As of today, AI seems to advance at a rhythm that outpaces current legislation. So one must remain ethically aware of these (and more) aspects when training, studying or deploying AI based products and services to the real world.



# Some important (pace) considerations

AI is a tool, a popular one. There are high hopes for what it can do and how much money there is to gain from it.

As of today, AI seems to advance at a rhythm that outpaces current legislation. So one must remain ethically aware of these (and more) aspects when training, studying or deploying AI based products and services to the real world.



**Dario Amodei**



## We Must Pace the Frontier

September 2026

Source: <https://darioamodei.com/post/we-must-pace-the-frontier>

# Ethical Considerations & Challenges



Algorithmic Bias & Fairness



Privacy & Data Security



Economic & Job Displacement\*



The "Black Box" Problem



Laws move slower than AI implementations / AI limitations are unclear regarding the law

# About the use of AI in the course

In general, it is accepted, but consider the following:

- It should not be used solely to build answers to your projects, without personal thought.
- It should be used to enhance your deliverables: include better visualizations, simplify your code, add features outside from what was requested (but that compliments your projects).
- Do not ask "please solve the project tasks listed in this document"
- Instead, you may ask:
  - "Put this project into executable tasks for a group of software Engineering students"
  - "List all libraries and documentation necessary to implement this project".
  - "I am having trouble understanding this step, please explaining it in simple terms and layout a implementation path for me to start solving the project"
- I recommend always adding this instruction: "make your your explanations do not give away direct answers to the project. Give clear instruction I can follow to implement the code instead of copy-and-past read code"

# About the use of AI in the course

Why does this matter?

License: CC BY-NC-ND 4.0  
arXiv:2601.20245v2 [cs.CY] 01 Feb 2026

## How AI Impacts Skill Formation

Judy Hanwen Shen

Alex Tamkin

August 24, 2026

Source: <https://arxiv.org/html/2601.20245v2>

## Rethinking Entry-Level Work and Talent Development in Age of AI

34 Pages • Posted: 11 Sep 2026

[Oluwaseyi Ayeni](#)

The University of Buckingham

[Chioma Ibukunola Egbochuku](#)

University of Hertfordshire

[juwon omigbodun](#)

Afe Babalola University

[Natasha Amaefuna](#)

University of Surrey

[Isabella Musinguzi-Karamukyo](#)

Imperial College London

[Francis Omigbodun](#)

Loughborough University

Source:

[https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=7447277](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7447277)

# Course details

15 weeks:

- Tuesdays and Wednesdays from 4pm to 5pm (subject to change according to professor's instructions)
- Evaluations:
  - 3 projects (80%): topics TBD
  - 1 lab (10%): Python for DA and ML
  - 1 essay (10%): AI and ethics
- Projects:
  - teams of 5 students (min), 6 at max
- **Assessments (base 20):**
  - **Code: 10 points**
    - **Delivery of the code based on the instructions given for each project.**
  - **Defense: 5 points**
    - **Oral comments given from students about their code.**
  - **Extension: 5 points**
    - **See next page**

# Course details

- **Assessments (base 20):**
  - **Code: 10 points**
  - **Defense: 5 points**
  - **Extension: 5 points**
    - This is a measure to mitigate the inflation of grades due to the use of AI.
    - You will certainly use GenAI, thus higher grades will be easier to achieve.
    - In the AI era, just asking for task execution is not enough.
    - Consequently, a quarter of your project grades will depend on your ability to go beyond what is asked.

**What is going “beyond”?**

**It strongly depends on creativity, and thus it is contradictory to constrict/define, but here are some hints:**

- 1. Model based: Exploring other algorithms/architectures to tackle the problem proposed (e.g. use more explainable models).**
- 2. Content based: Explore applications of the model you are implementing to solve different problems (e.g. take a model used to analyse market data and explore application in cybersecurity-related issues).**
- 3. Complexity based: make deeper analysis of the project and expanding your comprehension through metrics and visualizations.**

# Course details

- **Assessments (base 20):**
  - **Code: 10 points**
  - **Defense: 5 points**
  - **Extension: 5 points**

**Some criteria that will be used to assess your extensions?**

- 1. Functionality: whatever you do, it must run (without catastrophic errors)**
- 2. Theoretical comprehension: whatever you do, you must be able to explain it.**
- 3. Code comprehension: whatever you do, you must be able to explain how it related to the code.**
- 4. Reasonable connection: whatever you do, it must relate to the project in a reasonable manner.**
  - a. You cannot just code something different from what was asked in the project.**
  - b. It is an EXTENSION of the original project, not a second one unrelated to the first.**
  - c. If there are any doubts, you can always ask me before developing your code.**
  - d. In the absence of advice request, incorrect extensions will be unlikely tolerated.**

# Course details

## Relation with your professor:

- I am here to help you out, reach out to me whenever you want THROUGH EMAIL:
  - dramsaran [at] ucab [dot] edu [dot] ve
- You are given two-to-three weeks to deliver the code, I am always open to questions
  - It is ideal for you to develop your code and show me advances.
- I am not the only one responsible for your learning nor I can read your mind:
  - Raise your hand and ask questions.
  - Reach out.
  - Review class contents, do independent research.
  - Let me know if you need something (either individually or as a group), do not assume I can infer it.
- I recon AI and ML is not everyone's professional path:
  - if you do not like this subject, do your best to hold good student practices in order to get the most out of it.
  - if you like it, you can ask for professional advice inside and outside the classroom.
- Additional advices: explore topics before class, rest and eat well, try to relate what we learn with your interests.

# Course details

15 weeks:

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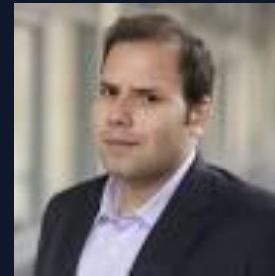
**Course team:**



**Lic. Didi Ramsaran-Chin**  
(me, psychologist)



**Dr. Pedro García**  
(physicist)



**Dr. Guillermo Palma**  
(engineering)



**MSc. Gustavo La Fontaine**  
(psychologist)

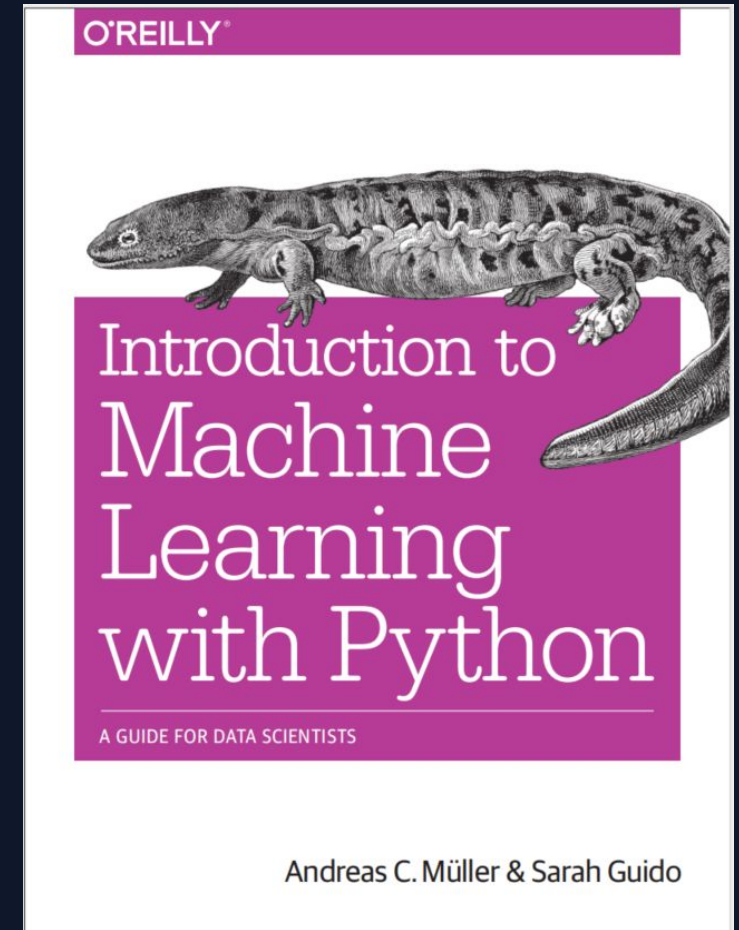
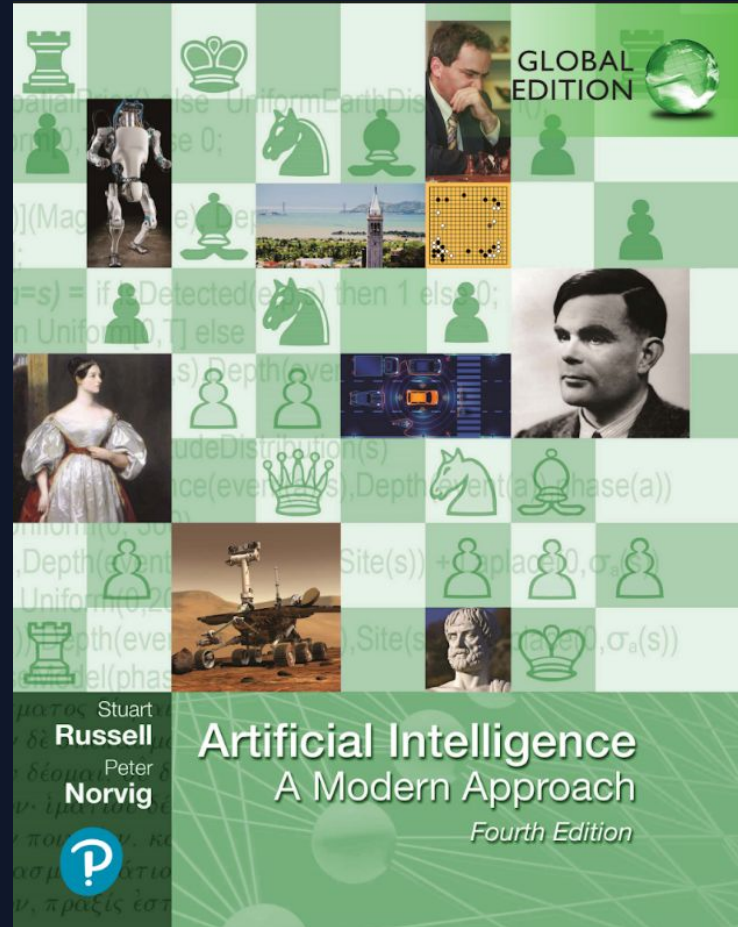
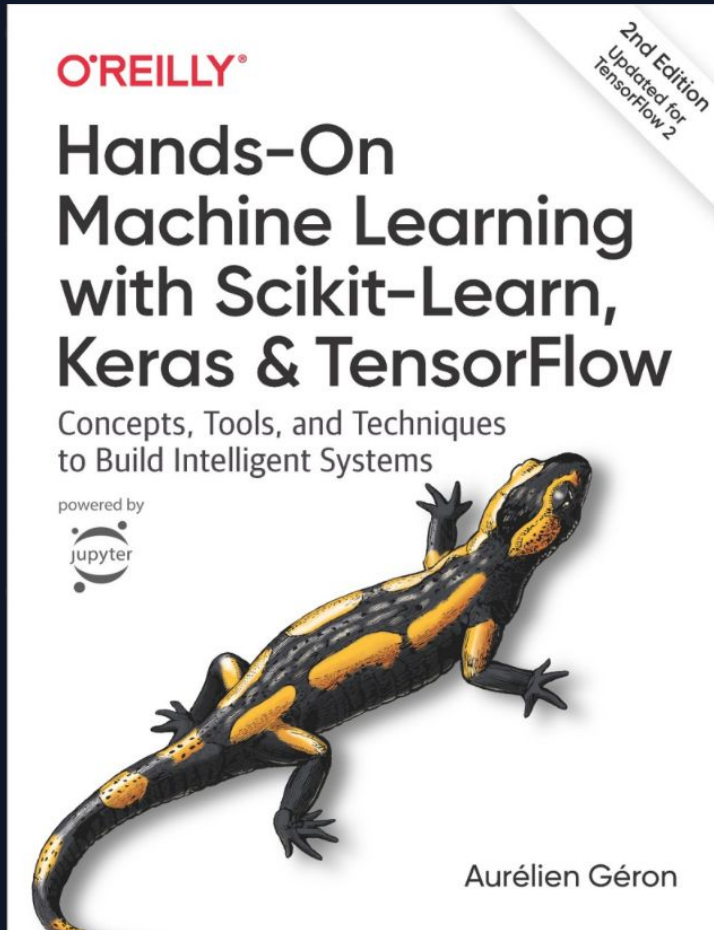
# Brief Bio

Lic. Didi Ramsaran-Chin  
(me, psychologist)

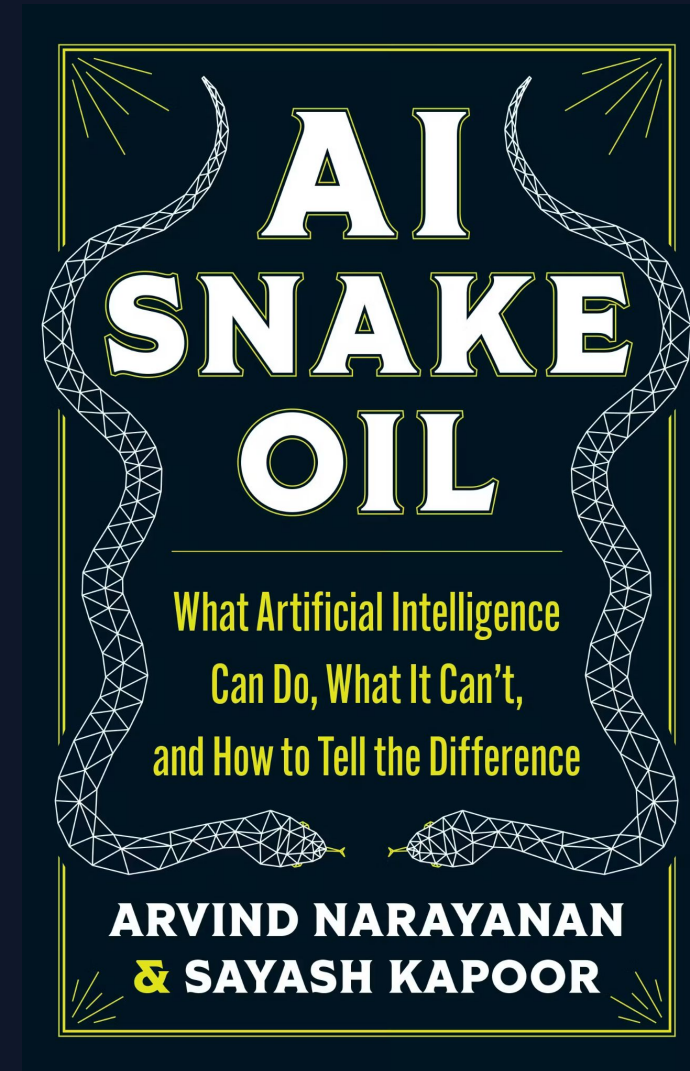
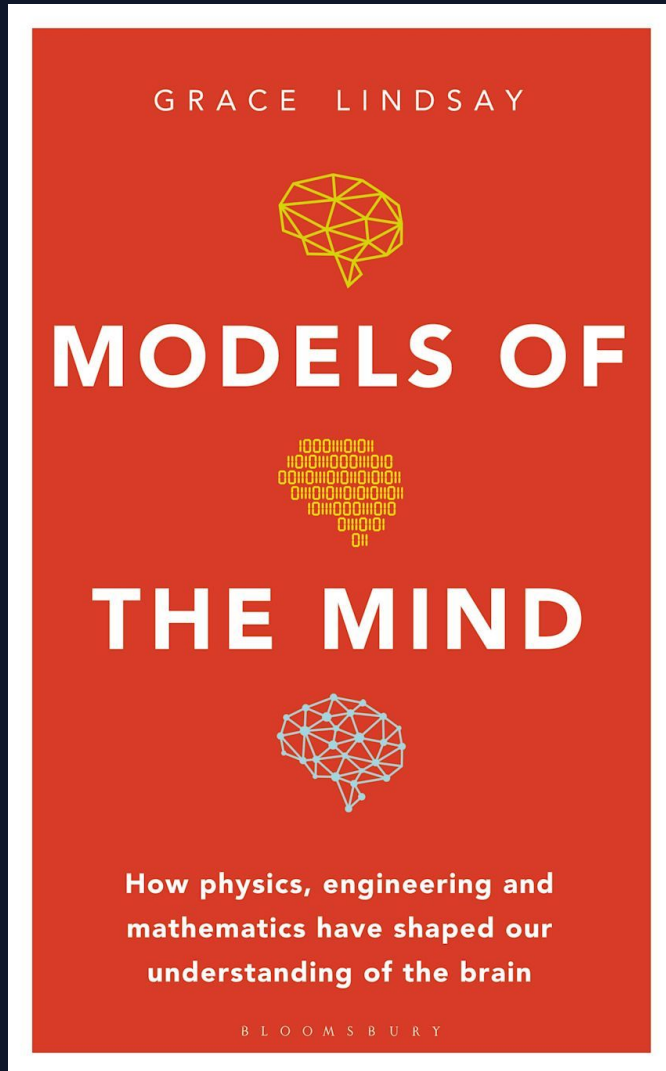


- Lic. Psychology (UCAB, 2025)
- Current MSc student in Systems modelling and simulation (Centre for Simulation and Models, Engineering, ULA)
- Current professor in AI and ML (Software Engineering, UCAB), and soon in AI and ethics (details to be shared).
- Aspiring scientist: Computational and theoretical Neuroscience, NeuroAI
- Research interest:
  - Complex systems methods applied to neural data analysis and modelling (i.e. Dynamical systems, algebraic topology, information theory).
  - Develop mathematical and computational models of (episodic) memory.
- Affiliations:
  - RedJIV-ACFIMAN [friends]
  - UCAB (Software engineering, Institute for the Study in Humanities), ULA (CESIMO)
  - Neuromatch

# Literature



# Literature recommendation



# References

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